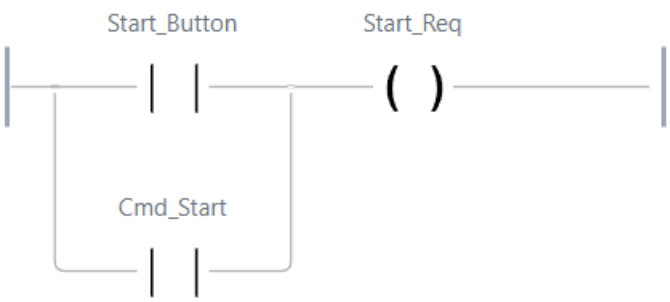
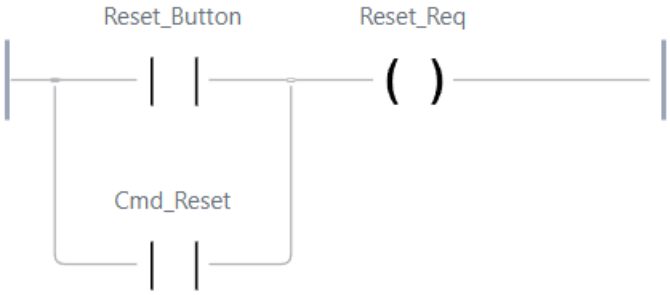
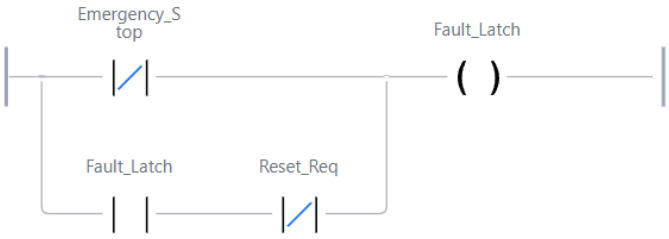
Start request: physical Start button OR dashboard Cmd_Start (Modbus coil 8192)



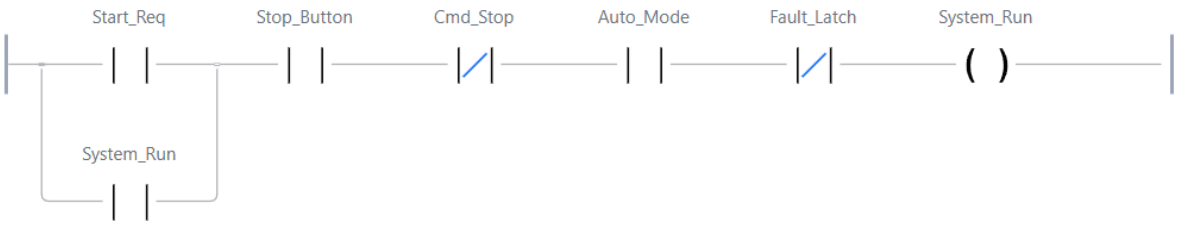
Reset request: physical Reset button OR dashboard Cmd_Reset (Modbus coil 8194)



Fault latch: Emergency stop pressed (NC input FALSE); cleared by Reset once released



System run: Start (button or dashboard) seal-in. Needs Stop OK (button and dashboard), Auto ON, no fault



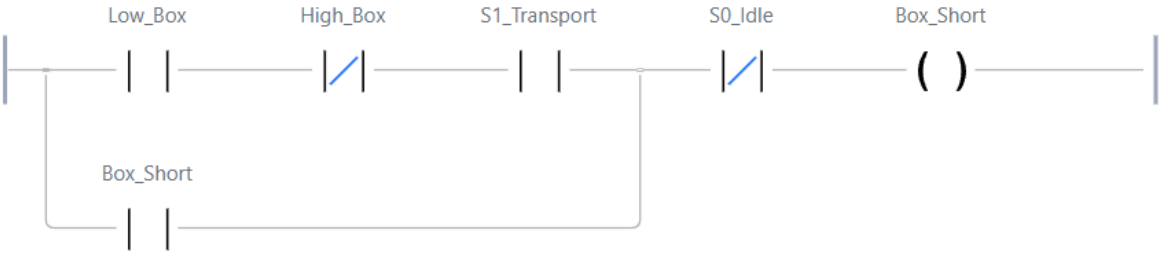
S0 Idle: no other step active (waiting for a box at At_Entry)



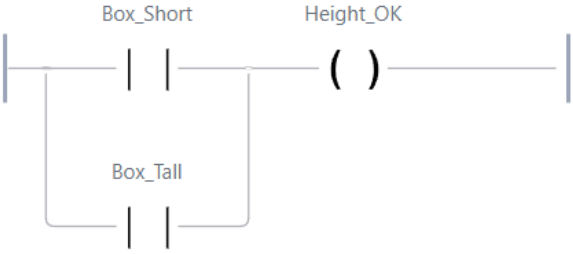
Height: tall box latched when High_Box sees it during transport; cleared when idle



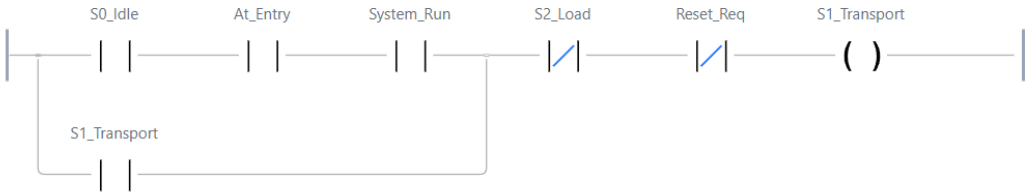
Height: short box latched when only Low_Box sees it during transport; cleared when idle



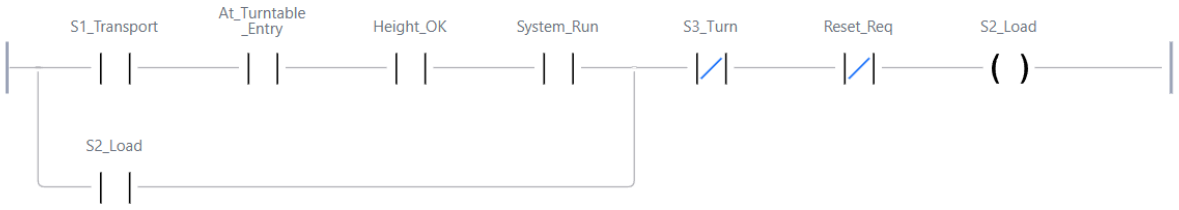
Height measured: short OR tall is known



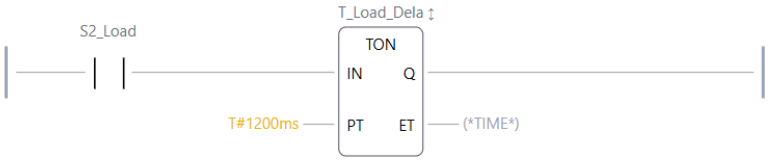
STEP 1 Transport: box at At_Entry, system idle -> feeder + entry conveyor move box past height sensors



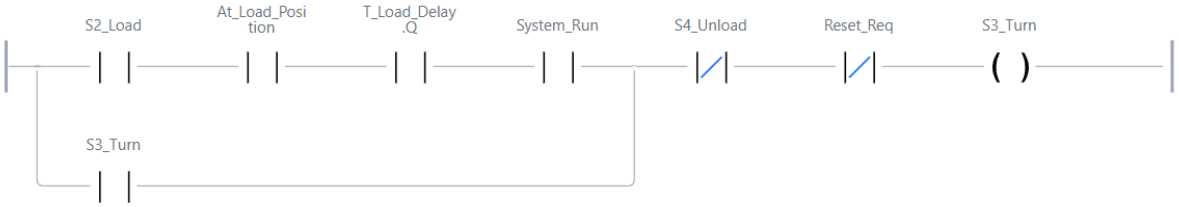
STEP 2 Load: box at turntable entry and height known -> load box onto turntable



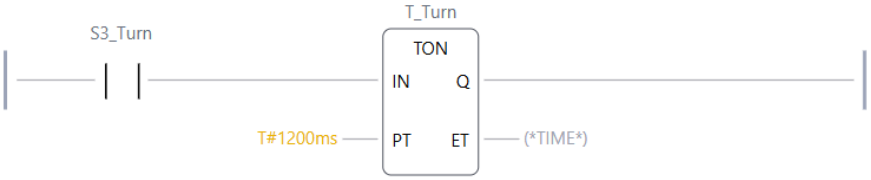
Load delay: 1.7 s after box detected at turntable entry, so it is fully on the turntable before turning



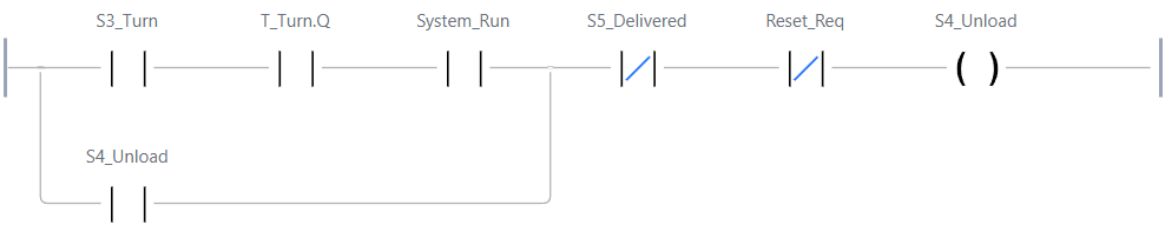
STEP 3 Turn: box at load position AND load delay done -> rotate turntable toward side conveyors



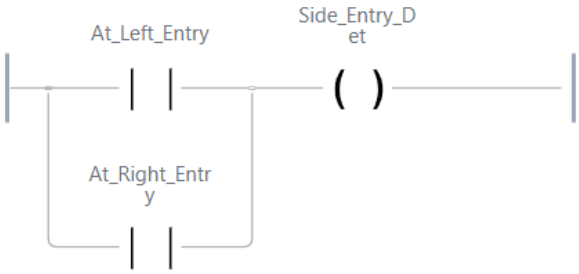
Turn wait: time for turntable to rotate (1.7 s)



STEP 4 Unload: turntable rotated -> push box to left (short) or right (tall)



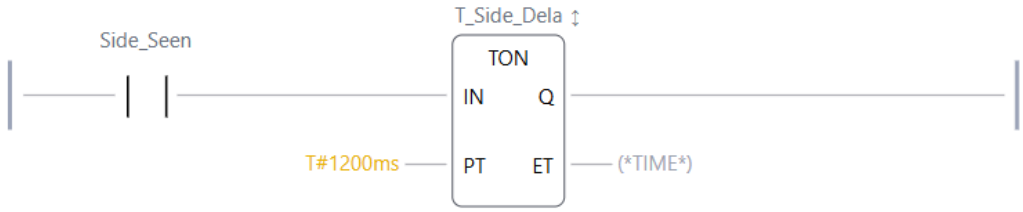
Box reached left OR right conveyor entry



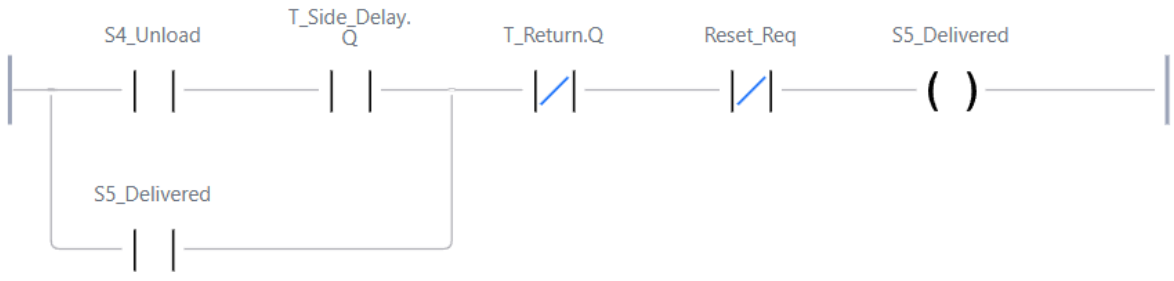
Side seen: latch that box reached left/right entry (kept even if sensor clears)



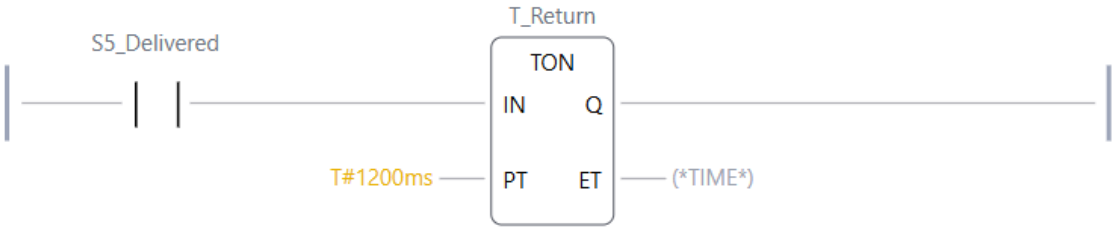
Side delay: 1.7 s after box detected at left/right entry, before turntable moves back



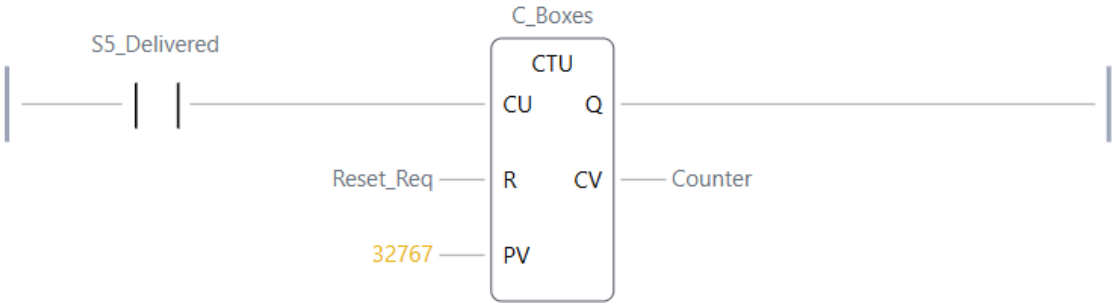
STEP 5 Delivered: side delay done -> counter +1, turntable returns; ends after T_Return



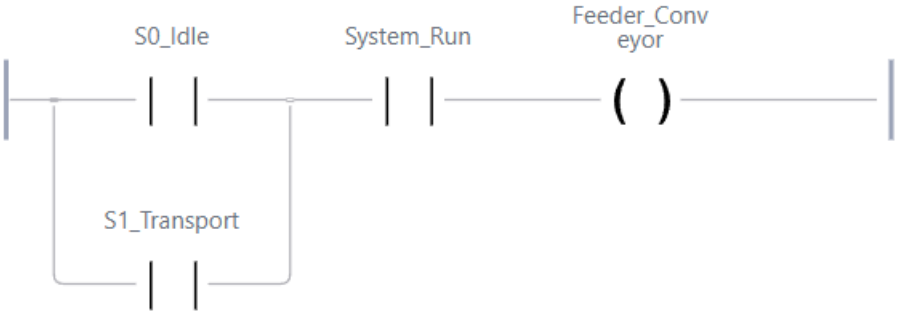
Return wait: time for turntable to rotate back (1.7 s)



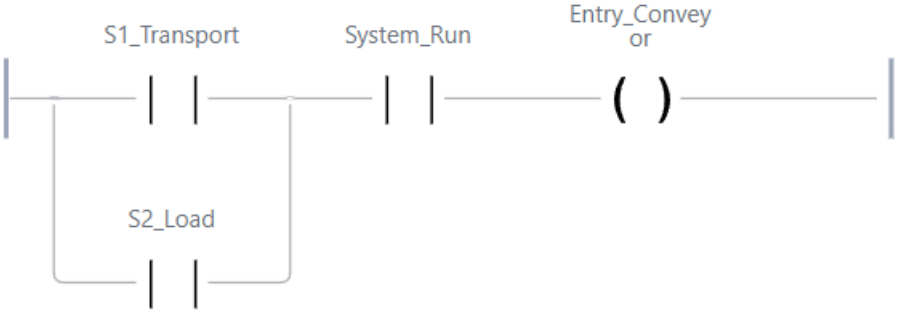
Counter: +1 for every box delivered (rising edge of S5); Reset clears; value -> Holding Reg 0



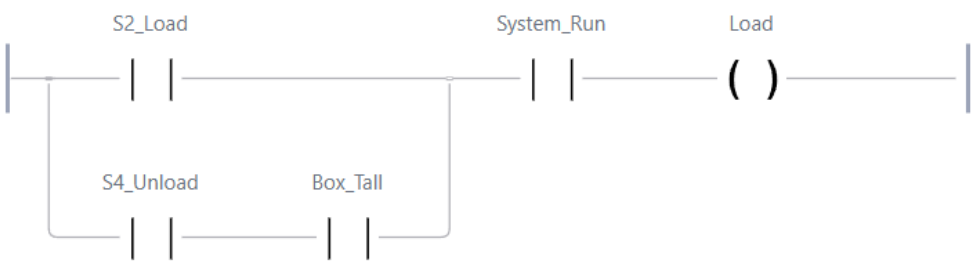
Feeder conveyor: waiting for box (S0) or transporting (S1)



Entry conveyor: transport (S1) or loading (S2)



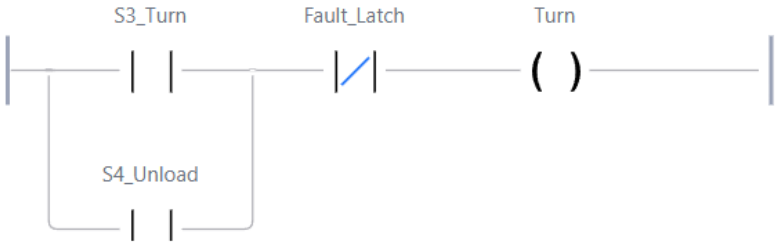
Load roller: loading (S2) OR unloading a TALL box (S4)



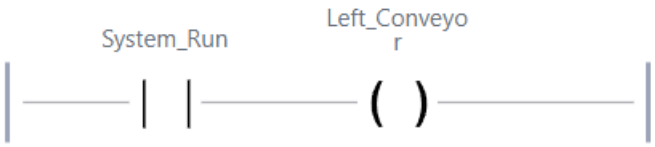
Unload roller: unloading a SHORT box (S4)



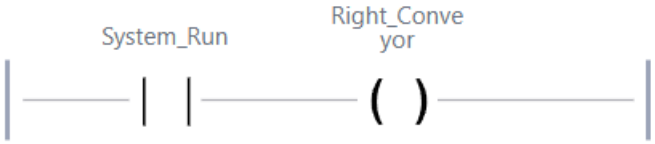
Turn: turntable rotated during S3 and S4 (off on fault)



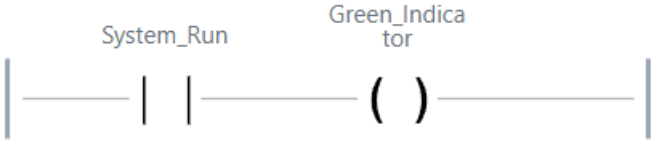
Left conveyor runs while system runs



Right conveyor runs while system runs



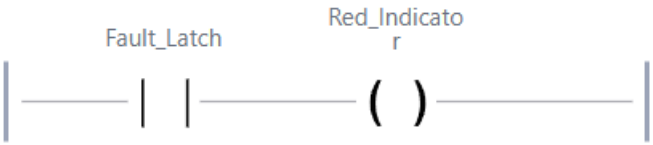
Green indicator: system running



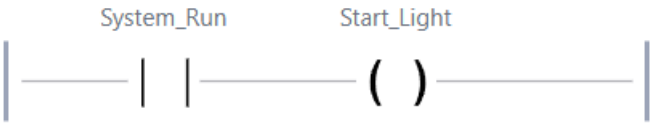
Yellow indicator: standby (stopped, no fault)



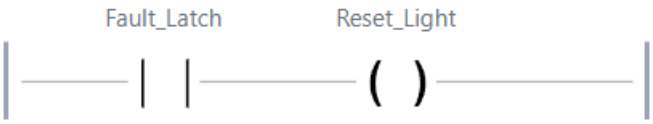
Red indicator: fault / emergency stop



Start light: running



Reset light: fault waiting for reset



Stop light: system not running

